

Solutions (Set B)

Q1) A user in Hyderabad, connected to the Internet via a 5 Mbps connection retrieves a 50 KB web page from a web server in Bangalore, where the page references 4 images of 300 KB size, each. Assume that the one way propagation delay is 20 ms. Approximately how long does it take for the page (including images) to appear on the user's screen, assuming non-persistent HTTP? (3 marks)

Sol: Total time for one request = 2 RTT + transmission time

Total number of objects = 5

RTT = 40 ms

Time for delivery of page = 2×40 + transmission time for 50 KB = 160 ms

Time for delivery of image = 2×40 + transmission time for 300 KB = 560 ms

Total time = 160 + 4x560 (ms) = 2.4 sec

Q2) Is it possible for an organization's web server and mail server to have exactly the same alias for a hostname (for example, foo.com)? What would be the type for the RR that contains the hostname of the mail server? (2 marks)

Sol: Yes, it is possible for an organization's web server and mail server to have exactly the same alias (hostname) like foo.com. The type for the Resource Record (RR) that contains the hostname of the mail server would be MX (Mail Exchange).

Q3) How does web caching reduce the delay in receiving a requested object? Will it be helpful in reducing the delay for all objects (including objects that are not cached) requested by a user or for only some of the objects? Why? (2 marks)

a) **Web caching reduces delay for cached objects** by serving them from a nearby location, avoiding long network transmission times and reducing server load.

b) **Only cached objects** benefit from the reduced delay. Non-cached objects must be retrieved from the origin server, and caching does not help reduce the delay for these initial requests.

Q4) Hosts A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 126. Suppose Host A then sends two segments to Host B back-to-back. The first and second segments contain 80 and 40 bytes of data, respectively. In the first segment, the sequence number is 127, the source port number is 302, and the destination port number is 80. Host B sends an acknowledgement whenever it receives a segment from Host A.

(0.5+0.5+1+2 marks)

a) In the second segment sent from Host A to B, what are the sequence number, source port number, and destination port number?

Sequence Number, Source Port, and Destination Port for the Second Segment

- **First Segment:**
 - Sequence number = 127
 - Number of bytes = 80
 - Source port number = 302
 - Destination port number = 80

The sequence number of the second segment will follow the bytes of the first segment.

- **Sequence number for the second segment:** The first segment contains 80 bytes, so the sequence number of the second segment will be:

$$127+80=207$$

Therefore, the second segment's **sequence number** is 207.

- **Source port number:** This remains the same as in the first segment, which is 302.
- **Destination port number:** This also remains the same, which is 80.

Answer (a):

- Sequence number = **207**
- Source port number = **302**
- Destination port number = **80**

(b) If the first segment arrives before the second segment, in the acknowledgement of the first arriving segment, what is the acknowledgement number, the source port number, and the destination port number?

Acknowledgement Number, Source Port, and Destination Port in the ACK of the First Segment

If the first segment (with sequence number 127) arrives before the second segment, Host B will send an acknowledgement.

- **Acknowledgement number:** The acknowledgement number indicates the next expected byte. Since Host B has received all bytes up to 126, and the first segment contains 80 bytes, the next expected byte will be:

$$127+80=207$$

Therefore, the **acknowledgement number** will be 207.

- **Source port number:** This will be the destination port number from the original segment, so it's **80** (the port used by Host B).

- **Destination port number:** This will be the source port number from the original segment, so it's **302** (the port used by Host A).

Answer (b):

- Acknowledgement number = **207**
- Source port number = **80**
- Destination port number = **302**

(c) If the second segment arrives before the first segment, in the acknowledgement of the first arriving segment, what is the acknowledgement number?

Acknowledgement Number if the Second Segment Arrives Before the First Segment

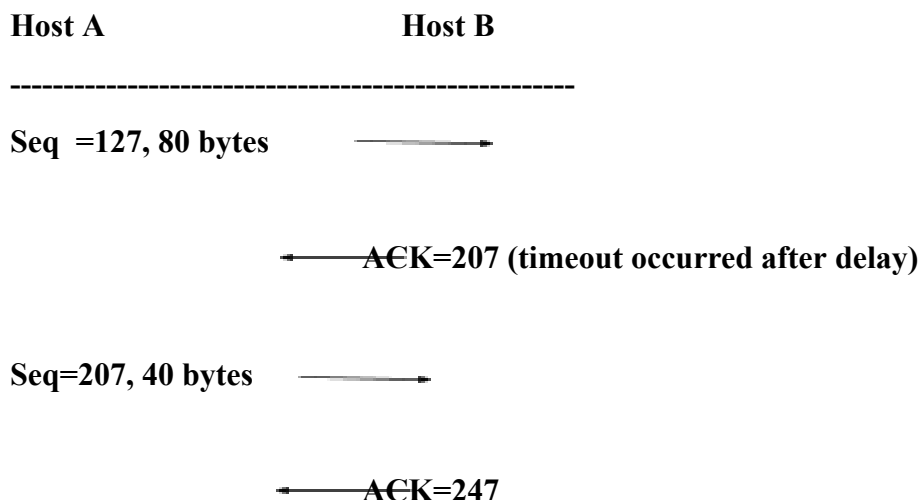
If the second segment (with sequence number 207) arrives first, Host B cannot acknowledge byte 207 yet because it hasn't received the earlier bytes (127-206). Therefore, Host B will acknowledge the last correctly received byte, which is 126.

- **Acknowledgement number:** The acknowledgement number will still be **127**, as Host B is still expecting the first byte of the first segment.

Answer (c): Acknowledgement number = **127**

(d) Suppose the two segments sent by A arrive in order at B. The first acknowledgement arrives after the first timeout interval. Draw the timing diagram, showing these segments and all other segments and acknowledgements sent. Assume that there is no additional packet loss. For each segment in your figure, provide the sequence number and number of bytes of data; for each acknowledgement that you add, provide the acknowledgement number.

Timing Diagram for Ordered Segment Arrival with Timeout and Acknowledgements



In this scenario, the two segments arrive in order, but the first acknowledgement is delayed due to a timeout. After that, the acknowledgement is sent, and there is no packet loss. Here's a step-by-step description:

1. Host A sends the first segment:
 - Sequence number = 127
 - Data = 80 bytes
2. Host A sends the second segment immediately after:
 - Sequence number = 207
 - Data = 40 bytes
3. Host B receives the first segment (seq 127–206) and expects the next byte to be 207, but it doesn't send an acknowledgement immediately due to the timeout.
4. Host B receives the second segment (seq 207–246). Since it now has all the data up to byte 246, it prepares an acknowledgement, but the first acknowledgement is delayed because the first timeout hasn't expired.
5. The first timeout on Host A expires. Host A retransmits the first segment.
6. Host B receives the retransmitted first segment, but it has already received that data, so it sends a cumulative acknowledgement for all data received up to byte 247 (since all data up to that point has been received).

Q5) Suppose a DNS resolver queries three DNS servers (A, B, and C) in sequence. Server A responds in 50 ms, Server B in 70 ms, and Server C in 30 ms. However, 30% of the time, Server A does not respond, requiring the resolver to query Server B next. Calculate the expected time it takes to resolve a DNS query. (3 marks)

Sol: Server A response time: 50 ms (70% success).

Server B response time: 50 ms (A fails) + 70 ms = 120 ms.

If Server C is not queried.

Expected time:

$$0.7 \times 50 \text{ ms} + 0.3 \times 120 \text{ ms} = 35 \text{ ms} + 36 \text{ ms} = 71 \text{ ms}$$

Partial marks have been given for those who considered, if Server C is queried.

Expected time = 70%(A)+30%(A+B+C)

$$= 0.7 \times 50 \text{ ms} + 0.3(50 \text{ ms} + 70 \text{ ms} + 30 \text{ ms}) = 35 + 45 = 80 \text{ ms}$$

Q6) Out of 12,000 UDP packets sent, 3% are randomly dropped due to network congestion. However, each lost packet is retransmitted, leading to a 5% increase in overall transmission due to overlap. Calculate the number of packets successfully delivered, taking into account the retransmitted packets. (3 marks)

Given Data:

- Total UDP packets sent = 12,000
- 3% of the packets are randomly dropped due to network congestion.
- Retransmission leads to a 5% increase in overall transmission due to overlap.

Step-by-Step Calculation:

1. Packets Dropped Initially:

3% of the 12,000 packets are dropped due to congestion.

Dropped packets = $0.03 \times 12,000 = 360$ packets

2. Retransmission and Overlap:

The retransmission of the dropped packets causes a 5% increase in the total transmission due to overlap. This means that the number of packets sent increases by 5% of the original total.

Increase due to retransmission = $0.05 \times 12,000 = 600$ packets

Thus, the total number of packets sent due to retransmissions and overlaps is:

Total packets sent after retransmissions = $12,000 + 600 = 12,600$ packets

3. Packets Successfully Delivered:

All the originally dropped packets (360 packets) are retransmitted and successfully delivered. Since no further packets are indicated to be dropped in the retransmission process, the total number of packets successfully delivered is the same as the total number of packets sent after retransmission.

Thus, the total number of successfully delivered packets is:

Successfully delivered packets = 12,600

Final Answer:

The number of packets successfully delivered, including retransmitted packets, is **12,600**.

7) In a Selective Repeat protocol, the maximum sequence number is 16, and the window size is 8. If one packet is lost every 16 packets, calculate the efficiency as a percentage. (3 marks)

SOL:

In the **Selective Repeat (SR)** protocol, efficiency depends on how many packets are successfully transmitted without being lost or requiring retransmission. Let's go through the calculation step by step, using the provided data.

Given Data:

- Maximum sequence number: 16
- Window size: 8
- One packet is lost every 16 packets.

Step-by-Step Calculation:

1. Selective Repeat Protocol Efficiency Overview

- In Selective Repeat, each packet is acknowledged individually. If a packet is lost, only that specific packet is retransmitted.
- The window size in Selective Repeat dictates how many packets can be sent before waiting for an acknowledgment (i.e., the sender can send up to 8 packets without waiting for ACKs).

2. Effect of Packet Loss

- One packet is lost for every 16 packets sent.
- In each group of 16 packets, only **15 packets are successfully delivered** without requiring retransmission.
- The lost packet must be retransmitted.

3. Transmission of 16 Packets

- The total number of packets sent to successfully deliver 16 packets includes the original 16 packets and the retransmission of the lost packet.
 - **16 packets are originally sent.**
 - **1 packet is lost** and retransmitted.
 - **Total packets sent = 16 + 1 = 17 packets** to deliver 16 packets.

4. Efficiency Calculation

Efficiency is the ratio of useful transmissions (i.e., the packets successfully delivered) to the total transmissions (including retransmissions).

$$\text{Efficiency} = (\text{Useful transmissions} / \text{Total transmissions}) * 100$$

In this case:

$$\text{Efficiency} = (16/17) * 100 = 94.12 \% \text{ (NB: There is no role of Window size in SR only the lost packet is resent)}$$

Final Answer:

The efficiency of the Selective Repeat protocol is **94.12%**.